MCV4U COMPREHENSIVE REVIEW PACKAGE

Vector & Parametric Equations, Planes in $\mathbb{R}^3$, Coplanarity, Distances & Line-Plane Geometry Complete Solutions Attached

1. Determine both the vector equation and the set of parametric equations for the plane π that passes through the point $A(2, 3, -1)$ and is parallel to the vectors $\vec{u} = (1, -2, 4)$ and $\vec{v} = (3, 0, -1)$.

2. Consider the plane $\pi : \vec{r} = (4, -1, 2) + s(2, 1, -3) + t(-1, 2, 5),\ s, t \in \mathbb{R}$.

a) Write the corresponding set of parametric equations for plane π.

b) Determine whether the point $Q(7, 3, -7)$ lies on plane π.

3. Determine a vector equation for the plane passing through $P(-2, 1, 5)$ and containing the line $L : \vec{r} = (1, -1, 3) + s(2, 4, -1),\ s \in \mathbb{R}$.

4. Determine the point where the plane $\pi : \vec{r} = (0, 2, -3) + s(1, 1, 2) + t(-2, 1, 1),\ s, t \in \mathbb{R}$ intersects the xy-coordinate plane ($z = 0$).

5. State with reasons whether the two planes $\pi_1 : \vec{r} = (1, 0, 2) + s(-1, 2, 1) + t(2, 1, 1)$ and $\pi_2 : \vec{r} = (3, 1, -1) + u(1, 1, -1) + v(3, 1, -2)$ intersect along a line, are parallel and distinct, or are coincident.

6. Express the vector $\vec{v} = (-5, 17)$ as a linear combination of $\vec{a} = (3, 1)$ and $\vec{b} = (-2, 4)$.

7. Determine whether the three 3D vectors $\vec{a} = (2, -1, 3)$, $\vec{b} = (1, 4, -2)$, and $\vec{c} = (3, 2, 1)$ are coplanar.

8. Explain geometrically why the vector $\vec{x} = (4, -2, 7)$ cannot be written as a linear combination of $\vec{y} = (1, 3, -2)$ and $\vec{z} = (-2, -6, 4)$.

9. Find the value of the scalar k such that the vectors $\vec{u} = (k, -4, 5)$, $\vec{v} = (2, 1, -3)$, and $\vec{w} = (1, 5, 3)$ are coplanar.

10. Prove that any vector $\vec{w} = (w_1, w_2)$ in $\mathbb{R}^2$ can be uniquely expressed as a linear combination of two non-zero, non-collinear vectors $\vec{u} = (u_1, u_2)$ and $\vec{v} = (v_1, v_2)$.

11. Determine the Cartesian (scalar) equation of the plane passing through $P_0(3, -2, 4)$ with normal vector $\vec{n} = (5, 1, -2)$.

12. Find the Cartesian equation of the plane containing the three non-collinear points $A(2, 1, 0)$, $B(4, -1, 3)$, and $C(1, 3, -2)$.

13. Convert the vector equation $\vec{r} = (1, -2, 3) + s(2, 1, -1) + t(-1, 3, 2),\ s, t \in \mathbb{R}$ into Cartesian form $Ax + By + Cz + D = 0$.

14. Given the plane $\pi : 3x - 4y + 2z - 12 = 0$:

a) Find the x-, y-, and z-intercepts of the plane.

b) Write a vector equation for the plane using its intercepts.

15. Determine the Cartesian equation of the plane that passes through $M(-1, 2, 5)$ and is perpendicular to the line $L : \frac{x-2}{2} = \frac{y+1}{-1} = \frac{z-4}{3}$.

16. Calculate the exact perpendicular distance from the point $P(1, 4, -2)$ to the plane $\pi : 3x - 4y + 7 = 0$.

17. Find the exact distance from the point $Q(2, -3, 1)$ to the 3D line $L : \vec{r} = (-1, 0, 2) + t(1, 2, -2),\ t \in \mathbb{R}$.

18. Calculate the exact distance between the point $P(-2, 5, 3)$ and the plane $\pi : 2x - y + 2z - 11 = 0$.

19. Find the exact distance between the two parallel planes $\pi_1 : 2x-y+2z-8=0$ and $\pi_2 : 4x-2y+4z+5=0$.

20. A 3D line L passes through $A(1,0,-1)$ with direction vector $\vec{m}=(2,-1,2)$. Find the exact coordinates of the point N lying on line L that is closest to the external point $P(4,2,1)$.

21. Calculate the acute angle θ (to the nearest degree) between the two intersecting planes $\pi_1 : 3x-2y+6z-1=0$ and $\pi_2 : x+y-4=0$.

22. Determine whether the line $L : \vec{r}=(2,1,-3)+t(1,-2,3),\ t\in\mathbb{R}$ intersects the plane $\pi : 2x+y-5=0$, lies completely within it, or is parallel and distinct.

23. Calculate the exact acute angle ϕ between the line $L : \vec{r}=(1,0,-2)+t(1,-2,2),\ t\in\mathbb{R}$ and the plane $\pi : 2x-y-z+4=0$.

24. Determine a parametric vector equation for the line of intersection between $\pi_1 : 2x+y-z=3$ and $\pi_2 : x-2y+2z=1$.

25. Solve the system of three planes algebraically using elimination and state the point of intersection $P(x,y,z)$:

$$\pi_1 : x+y+z=6$$
$$\pi_2 : 2x-y+z=2$$
$$\pi_3 : x+2y-z=7$$

26. Solve the system of three planes: $\pi_1 : x+y+z=6,\ \pi_2 : 2x-y+z=3,\ \pi_3 : 3x+2z=9$. Show that the three planes intersect along a common line and state its parametric vector equation.

MASTER EXEMPLAR SOLUTIONS

Step-by-Step Solutions for Questions 1–26

1. $\vec{r} = (2,3,-1) + s(1,-2,4) + t(3,0,-1),\ s,t \in \mathbb{R}$. Parametric: $x = 2+s+3t,\ y = 3-2s,\ z = -1+4s-t$.

2. a) $x = 4+2s-t,\ y = -1+s+2t,\ z = 2-3s+5t$. b) Sub $Q(7,3,-7)$: system yields $s = 2, t = 1 \implies z = 1 \neq -7$. Point Q does not lie on plane.

3. Line point $P_0(1,-1,3)$, external point $P(-2,1,5) \implies \vec{P_0P} = (-3,2,2)$. Equation: $\vec{r} = (-2,1,5) + s(2,4,-1) + t(-3,2,2),\ s,t \in \mathbb{R}$.

4. Parametric $x = s-2t,\ y = 2+s+t,\ z = -3+2s+t$. Set $z = 0 \implies t = 3-2s$. Sub t: $x = -6+5s,\ y = 5-s,\ z = 0,\ s \in \mathbb{R}$.

5. Normal $\vec{n}_1 = (-1,5,3),\ \vec{n}_2 = (2,-8,-6)$. Since $\vec{n}_1 \nparallel \vec{n}_2$, planes intersect along a line.

6. $a(3,1) + b(-2,4) = (-5,17) \implies 3a-2b = -5,\ a+4b = 17 \implies a = 1,\ b = 4$.

7. $\vec{a} \cdot (\vec{b} \times \vec{c}) = (2,-1,3) \cdot (10,-9,-13) = 20+9-39 = -10 \neq 0 \implies$ Not coplanar.

8. $\vec{z} = -2\vec{y}$, so $\vec{y}$ and $\vec{z}$ are collinear and span only a 1D line in $\mathbb{R}^3$. Since $\vec{x}$ is not collinear with $\vec{y}$, it cannot be formed as a linear combination.

9. $\vec{w} \cdot (\vec{u} \times \vec{v}) = 0 \implies (k,-4,5) \cdot (2,7,6) = 2k-28+30 = 0 \implies 2k+2 = 0 \implies k = -1$.

10. Since $\vec{u} \nparallel \vec{v}$, determinant $D = u_1v_2 - u_2v_1 \neq 0$. System $au_1 + bv_1 = w_1,\ au_2 + bv_2 = w_2$ has a unique solution for any (w_1, w_2) by Cramer's Rule.

11. $5(x-3) + 1(y+2) - 2(z-4) = 0 \implies 5x+y-2z-5 = 0$.

12. $\vec{AB} = (2,-2,3),\ \vec{AC} = (-1,2,-2) \implies \vec{n} = (-2,1,2) \sim (2,-1,-2)$. Sub $A(2,1,0) \implies 2(2) - 1(1) - 2(0) + D = 0 \implies D = -3 \implies 2x-y-2z-3 = 0$.

13. Eliminate s,t: $\vec{n} = (2,1,-1) \times (-1,3,2) = (5,-3,7)$. Sub $(1,-2,3) \implies D = -32 \implies 5x-3y+7z-32 = 0$.

14. a) x-int $(4,0,0)$, y-int $(0,-3,0)$, z-int $(0,0,6)$. b) $\vec{d_1} = (-4,-3,0),\ \vec{d_2} = (-4,0,6) \implies \vec{r} = (4,0,0) + s(-4,-3,0) + t(-4,0,6)$.

15. Normal $\vec{n} = (2,-1,3)$. Sub $M(-1,2,5) \implies 2(-1) - 1(2) + 3(5) + D = 0 \implies D = -11 \implies 2x-y+3z-11 = 0$.

16. $d = \frac{|3(1)-4(4)+7|}{\sqrt{3^2+(-4)^2}} = \frac{|3-16+7|}{5} = \frac{6}{5} = 1.2$ units.

17. $P_0(-1,0,2),\ \vec{m} = (1,2,-2),\ \vec{P_0Q} = (3,-3,-1) \implies \vec{P_0Q} \times \vec{m} = (8,5,9)$. Distance $d = \frac{\sqrt{8^2+5^2+9^2}}{\sqrt{1^2+2^2+(-2)^2}} = \frac{\sqrt{170}}{3}$ units.

18. $d = \frac{|2(-2)-1(5)+2(3)-11|}{\sqrt{2^2+(-1)^2+2^2}} = \frac{|-4-5+6-11|}{3} = \frac{14}{3}$ units.

19. Scale $\pi_2 : 4x-2y+4z-16 = 0$. Pick $P_0(0,0,4)$ on $\pi_2 \implies d = \frac{|4(0)-2(0)+4(4)+5|}{\sqrt{16+4+16}} = \frac{21}{6} = 3.5$ units.

20. $N(1+2t,-t,-1+2t) \implies \vec{PN} = (2t-3,-t-2,2t-2)$. Set $\vec{PN} \cdot \vec{m} = 0 \implies 2(2t-3) - 1(-t-2) + 2(2t-2) = 0 \implies 9t-8 = 0 \implies t = \frac{8}{9} \implies N\left(\frac{25}{9}, -\frac{8}{9}, \frac{7}{9}\right)$.

21. $\vec{n}_1 = (3,-2,6),\ \vec{n}_2 = (1,1,0) \implies \cos\theta = \frac{|3(1)-2(1)+0|}{7\sqrt{2}} = \frac{1}{7\sqrt{2}} \implies \theta \approx 84°$.

22. $\vec{d}\cdot\vec{n} = (1,-2,3)\cdot(2,1,0) = 0$. Test point $(2,1,-3) : 2(2)+1-5=0 \implies$ Line lies completely within plane.

23. $\vec{d}\cdot\vec{n} = 2(1)-1(2)+2(-1) = -2$. $\sin\phi = \frac{|-2|}{3\sqrt{6}} \implies \phi \approx 16°$.

24. Direction $\vec{m} = \vec{n}_1 \times \vec{n}_2 = (5,-5,-5) \sim (1,-1,-1)$. Point on line: set $z=0 \implies x=1, y=1 \implies x = 1+t, y = 1-t, z = -t$.

25. (1) $x+y+z=6$, (2) $2x-y+z=2$, (3) $x+2y-z=7$. Add (1) + (3): $2x+3y=13$. Add (2) + (3): $3x+y=9 \implies y=9-3x$. Sub into $2x+3(9-3x)=13 \implies -7x=-14 \implies x=2$. Then $y=3, z=1$. Unique POI $P(2,3,1)$.

26. $\pi_1+\pi_2$ gives $3x+2z=9$ (π_3). The planes intersect along a common line.